

Example 2: Fetch data from Statistics Estonia with R

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Setup R

```
1 library(quarto) # for compiling Quarto presentations and documents
2 library(pxweb) # query and fetch data from Statistics Estonia
3 library(tidyverse) # dplyr, ggplot2, readr, etc.
4 library(here) # path settings
```

Accessing Statistics Estonia Data with **pxweb**

- **Statistics Estonia** is not yet part of DBnomics
- There is **no** dedicated R package for andmed.stat.ee
- **However**, you can access the data using the [pxweb](#) R package
- It is based on the **PX-WEB API**, also used by many Nordic statistical agencies

Data Access Options

- **Interactive Access**
Browse tables and build queries via the Statistics Estonia website in the R console
- **Direct Programmatic Query**
Use the API endpoint (a *URL*) and supply **key-value pairs** to specify:
 - Variables (e.g., indicators, time, sector)
 - Values or selections for each variable

Find and download data interactively

```
1 pxweb_interactive() # All available databases
2 pxweb_interactive("https://andmed.stat.ee/api/v1/en/stat") # Statistics Estonia
```

```
1 #| eval: false
2 =====
3 R PXWEB: Content of 'andmed.stat.ee'
4       at '/api/v1/en/stat'
5 =====
6 [ 1 ] : Environment
7 [ 2 ] : Economy
8 [ 3 ] : Population
9 [ 4 ] : Social life
10 [ 5 ] : Multidomain statistics
11 [ 6 ] : Population and Housing Census
12 [ 7 ] : Discontinued datasets
13 =====
```

```
14 Enter your choice:
15 ('esc' = Quit, 'b' = Back, 'i' = Show id)
16 1:
```

GET call to PXWEB API

- Example dataset: PA111 - AVERAGE MONTHLY GROSS WAGES (SALARIES)

```
1 pa111_meta <- pxweb_get(  
2   url = "https://andmed.stat.ee/api/v1/en/stat/PA111")  
3 pa111_meta
```

PXWEB METADATA

PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES BY ECONOMIC ACTIVITY SECTION (QUARTERLY)

variables:

```
[[1]] Näitaja: Indicator  
[[2]] Tegevusala: Economic activity  
[[3]] Vaatlusperiood: Reference period
```

The object is just metadata, no data yet!

How to identify key-value pairs?

- Use either:
 1. Programmatically via the metadata object
 2. From the Statistics Estonia web interface – click “API query for this table”

Key-value pairs

- Values for the first key (Näitaja)

```
1 pa111_meta$variables[[1]]
```

```
$code
```

```
[1] "Näitaja"
```

```
$text
```

```
[1] "Indicator"
```

```
$values
```

```
[1] "GR_W_AVG"      "NR_EMPL"      "GR_W_D1"      "GR_W_D2"      "GR_W_D3"  
[6] "GR_W_D4"      "GR_W_D5"      "GR_W_D6"      "GR_W_D7"      "GR_W_D8"  
[11] "GR_W_D9"      "GR_W_AVG_SM"
```

```
$valueTexts
```

```
[1] "Average monthly gross wages (salaries), euros"  
[2] "Number of employees"  
[3] "1st decile of monthly gross wages (salaries), euros"  
[4] "2nd decile of monthly gross wages (salaries), euros"  
[5] "3rd decile of monthly gross wages (salaries), euros"  
[6] "4th decile of monthly gross wages (salaries), euros"  
[7] "Median (5th decile) of monthly gross wages (salaries), euros"  
[8] "6th decile of monthly gross wages (salaries), euros"
```

Fetch data

- Set: Key-value pair (as R list object) & Url to the database
- Wildcard operator `*` to select all values

```
1 px_query_list1 <- list(  
2   "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),  
3   "Tegevusala" = "*",  
4   "Vaatlusperiood" = "*" )  
5  
6  
7 pa111 <- pxweb_get(  
8   url = "https://andmed.stat.ee/api/v1/en/stat/PA111",  
9   query = px_query_list1  
10 )
```

- We get a list object with the data

```
1 pa111
```

PXWEB DATA

With 4 variables and 960 observations.

```
1 class(pa111)
```

```
[1] "pxweb_data" "list"
```


Convert the pxweb list object to a data frame

```
1 # Convert to data frame
2 pa111.df <- as.data.frame(pa111, column.name.type = "text", variable.value.type = "text")
3 pa111.code <- as.data.frame(pa111, column.name.type = "code", variable.value.type = "code")
4 # Optional: Combine text and code in one data frame
5 pa111.df$Näitaja <- pa111.code$Näitaja
```

```
1 head(pa111.df, n=3)
```

	Indicator	Economic activity
1	1st decile of monthly gross wages (salaries), euros Total	all activities
2	1st decile of monthly gross wages (salaries), euros Total	all activities
3	1st decile of monthly gross wages (salaries), euros Total	all activities
	Reference period	
1	2021 Q1	
2	2021 Q2	
3	2021 Q3	
	PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES	
1		532
2		472
3		508
	Näitaja	
1	GR_W_D1	
2	GR_W_D1	
3	GR_W_D1	

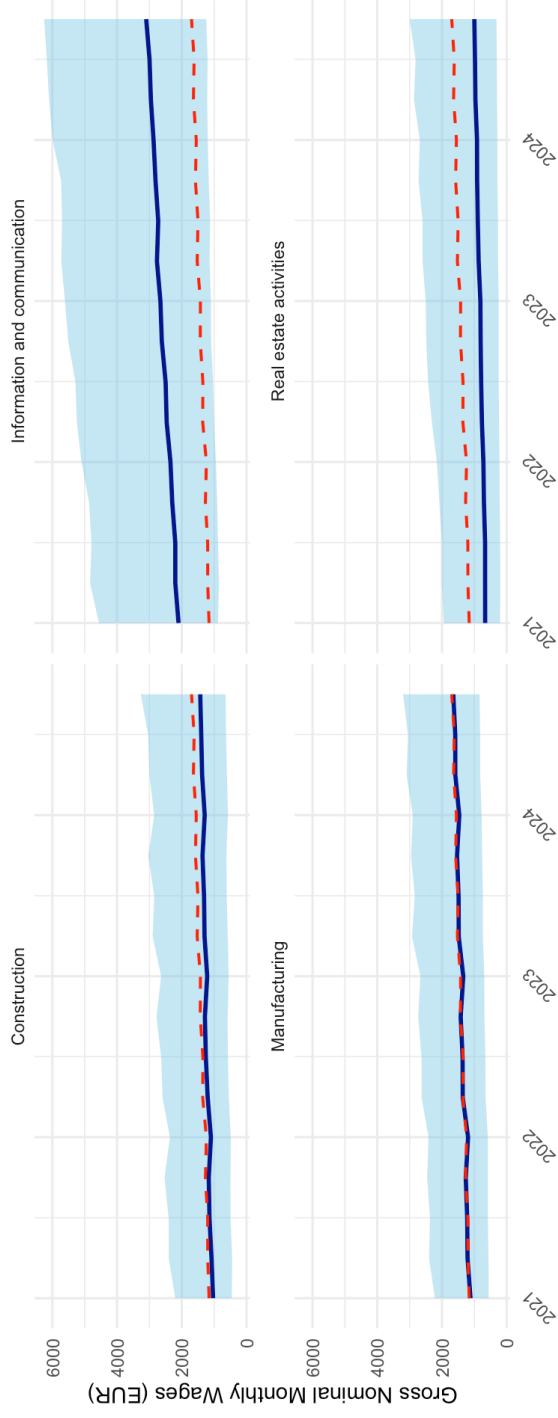
```

1 ggplot(pa111_plotdata, aes(x = quarter)) +
2   geom_ribbon(aes(ymax = D1, ymin = D9), fill = "skyblue", alpha = 0.5) +
3   geom_line(aes(y = Median), color = "darkblue", linewidth = 1) +
4   geom_line(aes(y = total_median), color = "red", linetype = "dashed", linewidth = 0.7) +
5   facet_wrap(~`Economic activity`, ncol = 2) +
6   labs(
7     title = "Wage Distribution by Sector (1st to 9th Decile, Median)",
8     subtitle = "Red dashed line shows overall median (Total - all activities)",
9     caption = source_note,
10    x = "",
11    y = "Gross Nominal Monthly Wages (EUR)"
12  ) +
13  theme_minimal() +
14  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Wage Distribution by Sector (1st to 9th Decile, Median)

Red dashed line shows overall median (Total - all activities)



"PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DECILES AND NUMBER OF EMPLOYEES by Indicator, Economic activity and Reference period"
Source: Statistics Estonia

Rendering All Formats from R or Bash

To render the document to **all** output formats (docx, Beamer, Reveal.js): run programmatically from another script or the R console:

```
1 quarto::quarto_render(here("teaching_materials", "example2_statistics_estonia.qmd"), output_format = "all")
```

or from the command line:

```
1 quarto render example2_statistics_estonia.qmd
```

End of session